Multimodal Fake vs Real Job Posting Classification

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Abstract

This project develops an explainable multimodal fraud detection system that combines text, image, and structured metadata to distinguish legitimate and fraudulent job postings. Using DistilBERT for text encoding, ResNet-50 for image analysis, and a metadata MLP for categorical features, we build a unified fusion architecture trained with weighted BCE loss to counter strong class imbalance. Three XAI techniques—SHAP, Integrated Gradients, and Grad-CAM—are integrated directly into the system to provide token-level, pixel-level, and modality-level interpretability. We demonstrate that multimodality improves robustness and that XAI reveals complementary insights such as vague descriptions, missing details, weak company logos, and incomplete profiles that contribute to fraud predictions. This work addresses a core need for transparent, auditable AI systems in online recruitment fraud prevention.

Keywords

Explainable AI, Multimodal Learning, Fraud Detection, SHAP, Integrated Gradients, Grad-CAM

1 Introduction

Online recruitment platforms are increasingly targeted by fraudulent job postings that attempt to harvest personal data, solicit payments, or mislead job seekers. The Fake Job Postings Dataset used in this project contains nearly 18,000 entries, with fraud representing only about 4.8%—creating severe class imbalance that complicates detection.

Traditional systems rely primarily on textual cues or handcrafted features, but modern fraudulent listings blend text, metadata, and visual elements (e.g., logos). To address this, we propose a multimodal deep learning system integrating text, structured metadata, and images.

In addition to accurate prediction, fraud detection requires transparency. Black-box models are difficult to trust or audit, especially in high-risk domains. Our system integrates Explainable AI (XAI) throughout the pipeline using SHAP, Integrated Gradients, and Grad-CAM to provide human-interpretable explanations.

2 Important Definitions and Problem Statement

2.1 Key Definitions

Fake Job Posting A job advertisement intentionally crafted to mislead applicants. These postings often mimic the structure or branding of legitimate listings but are designed to extract sensitive personal information, solicit payments, or direct users to malicious websites. Their subtlety makes manual detection difficult, highlighting the need for data-driven solutions.

Multimodal Learning A learning paradigm that integrates multiple sources of information, such as text, structured tabular data, and images into a unified predictive framework. In this project, multimodality allows the system to capture linguistic cues, hiring pattern irregularities, and visual inconsistencies associated with fraudulent job listings.

TF-IDF (Term Frequency–Inverse Document Frequency) A classic text-representation technique that quantifies how important a word is relative to both its document and the overall corpus. It supports traditional machine learning models by emphasizing terms that regularly appear in fraudulent postings.

BERT Embeddings Dense contextual embeddings generated using a pretrained BERT model. By capturing subtle semantic relationships and contextual meaning across job descriptions, requirements, and profiles, these embeddings serve as a strong foundation for the textual component of the classifier.

Convolutional Neural Network (CNN) A deep neural architecture specialized for extracting features from visual data. CNNs identify patterns such as shapes, textures, and layout anomalies, making them suitable for identifying inconsistencies in company logos, job-post images, or other visual elements related to a listing.

2.2 Problem Statement

Online recruitment platforms face a significant and growing challenge: distinguishing legitimate job postings from fraudulent ones. With a dataset of nearly 18,000 postings of which only about 4.8% are fraudulent, a simple heuristic or rule based approach quickly breaks down. Fraudulent listings often blend convincingly with genuine ones, and the imbalance between real and fake posts makes the minority class especially hard to detect.

This project aims to develop a multimodal classification system capable of accurately identifying fake job postings. The system draws on three complementary modalities:

- Unstructured text, capturing suspicious linguistic patterns or unrealistic role descriptions
- Structured categorical metadata, identifying atypical combinations of employment type, experience levels, education requirements, and industry markers
- Images, assessing potential visual inconsistencies such as questionable logos or formatting cues

By combining these sources of evidence, the goal is to create a robust model that improves fraud detection accuracy and contributes to safer online hiring ecosystems.

3 Overview of Proposed Approach / System

This work presents a multimodal learning framework that integrates information from three heterogeneous modalities: *text*, *categorical metadata*, and *image data*. Each modality captures a distinct aspect of the underlying context, and their joint modeling enables more robust and context-aware predictions.

The overall system follows a multi-stage multimodal fusion pipeline:

(1) **Modality-specific encoders** independently transform raw text, categorical variables, and images into dense semantic embeddings.
(2) **Embedding alignment and projection** place all modality representations into a compatible latent space.
(3) **Fusion module** combines the multimodal embeddings into a unified representation.
(4) **Downstream classifier** generates the final prediction from the fused vector.

The framework is designed to remain flexible and modular, allowing the integration of multiple fusion strategies such as feature-level fusion, late/post-hoc fusion, and conditional fusion mechanisms. Different team members explored these strategies to analyze how multimodal interactions influence model behavior and performance.

To enhance interpretability, the system incorporates an **explainability (XAI) layer**. Using gradient-based, attention-based, and attribution-based techniques, the model highlights modality-specific evidence that drives each prediction. These explanations help validate model reliability and provide insights into the interplay between textual, categorical, and visual cues.

The proposed system is designed to detect fraudulent job postings by integrating multiple data modalities—textual descriptions, structured metadata, and image content—into a unified predictive pipeline. The overall approach combines comprehensive exploratory data analysis, robust preprocessing strategies, and a suite of baseline and deep learning models, culminating in a multimodal fusion architecture.

The development process involved:

- Conducting extensive data exploration to understand distributional properties, missing data, and class imbalance.
- Preprocessing textual, categorical, and visual inputs using modality-specific techniques.
- Establishing strong baseline models for text and structured features.
- Developing and evaluating CNN-based models for image analysis.
- Integrating all three modalities into a cohesive multimodal framework with built-in interpretability mechanisms.

This coordinated pipeline provides a scalable foundation for accurate and explainable fraud detection across heterogeneous data sources.

3.1 Project Overview

- **Goal:** Build a predictive and interpretable system that distinguishes real vs. fraudulent job postings.
- **Dataset:** Fake Job Postings dataset (17,880 samples).
- **Modalities:**
  - Text (7 combined fields)
  - Structured metadata (employment type, experience, education, industry, function)
  - Images (company logos)
- **Challenges:**
  - Severe class imbalance
  - Missing values in metadata and text columns
  - Textual ambiguity and deceptive phrasing in fraud cases

3.2 Checkpoint Summary

Checkpoint 1: This phase focused on exploratory data analysis and initial text-based modeling. Key accomplishments included:

- Identifying the severe class imbalance in the dataset and adopting stratified splits for all experiments.
- Implementing TF–IDF baselines and the Complement Naïve Bayes classifier.
- Training an initial BERT-based text classifier, which significantly outperformed traditional models.
- Analyzing correlations of binary features such as *telecommuting*, *has_questions*, and *has_company_logo*.

Checkpoint 2: This phase expanded the project into full multimodality and deeper modeling. Key work included:

- Building the multimodal fusion pipeline integrating text, categorical metadata, and images.
- Benchmarking multiple CNN architectures, including VGG16, ResNet-50, MobileNetV2, and AlexNet.
- Engineering and encoding categorical metadata, including experience, education, industry, and salary fields.
- Conducting early multimodal fusion experiments combining structured, textual, and visual embeddings.

4 Technical Details of Proposed Approaches / Systems

4.1 4.1 Data Processing Pipeline

Text Processing:

- Combined textual fields (title, description, requirements, etc.)
- Tokenized with DistilBERT tokenizer (max 128 tokens)
- TF-IDF used for baselines

Metadata Processing:

- Label encoding for multimodal neural model
- One-hot encoding for NB/TF-IDF
- Salary parsing into salary_min, salary_max, salary_mean, and salary_missing

Image Processing:

- Resized to 224×224
- Normalized to ImageNet statistics
- Missing images replaced with zero-tensors

4.2 4.2 Multimodal Fusion Architecture

The architecture integrates three encoders:

```
Text Encoder (DistilBERT)   -> 768-d
Image Encoder (ResNet50)    -> 2048-d
Metadata MLP (4->64->64)    -> 64-d
```

```
Concatenate -> Dense(256) -> Dropout -> Dense(1)
```

Training uses:

- Weighted BCEWithLogits
- AdamW optimizer
- Batch size 8–32 depending on GPU memory
- Stratified train-test splits

4.3 4.3 CNN Backbone Experiments

A series of convolutional neural network (CNN) architectures were evaluated to establish strong image-based baselines. The experiments included both lightweight and deep models to assess performance across different capacity levels:

- **VGG16:** Utilized with light data augmentation and a frozen feature extractor. Served as a stable baseline and provided consistent performance under ROC–AUC monitoring.
- **MobileNetV2:** Delivered strong results with efficient computation, achieving high ROC–AUC and demonstrating suitability for deployment in resource-constrained settings.
- **ResNet50:** Fine-tuning of the final convolutional blocks contributed to robust generalization, making it one of the strongest image-based performers.
- **AlexNet:** Used primarily as a historical baseline; performance was lower compared to modern architectures but useful for comparative evaluation.

4.4 4.4 Integrated Explainability (XAI)

XAI is woven into the training/evaluation pipeline:

SHAP:

- Text: Token attribution via masking-based SHAP
- Image: Inpainting-based SHAP for pixel attributions

Integrated Gradients:

- Computes gradients w.r.t. text embeddings
- Zero embedding baseline

Grad-CAM:

- Applied to ResNet50 final conv block
- Highlights discriminative visual regions in logos

Modality Contribution:

$$\text{Contribution}_{\text{modality}} = P(\text{fraud}|\text{all}) - P(\text{fraud}|\text{modality ablated})$$

5 Experiments

5.1 5.1 Baseline Models

Complement Naive Bayes (TF-IDF + OHE): Moderate ROC-AUC, interpretable tokens.

Lasso Regression: Better ROC-AUC, strong feature selection properties.

BERT Text Classifier: State-of-the-art performance, achieving ROC-AUC ≈ 0.99.

5.2 5.2 CNN Image Models (Checkpoint 2)

- VGG16: Fast convergence, high PR-AUC
- MobileNetV2: Best lightweight model
- ResNet50: Best overall CNN performance
- AlexNet: Lower performance, used as baseline

5.3 5.3 Multimodal Fusion Model

Early runs show:

- Text dominates prediction early in training
- Images add discriminative cues (logo presence/quality)
- Metadata provides useful secondary signals

5.4 5.4 XAI Experimental Outputs

The system generates:

- Token-level SHAP heatmaps
- Image SHAP overlays
- ResNet50 Grad-CAM heatmaps
- JSON files with modality ablation scores

5.5 5.5 Faithfulness Evaluation

Token removal tests show consistent probability drop for top-K influential tokens.

6 Results and Analysis

- BERT significantly outperforms classic models
- ResNet50 outperforms other CNNs
- Multimodal fusion (text + image + metadata) increases robustness
- XAI reveals clear patterns:
  - Fraudulent postings contain vague, generic text
  - Missing or low-quality logos correlate with fraud
  - Metadata signals (salary missing, profile incomplete) reinforce fraud predictions

7 Related Work

Research on detecting fake job advertisements and fraudulent online recruitment has grown significantly. Below are a number of representative studies, organized by their methodological approaches and relevance to multimodal or machine-learning based detection.

7.1 Machine Learning and Traditional Approaches to Fake Job Detection

- Real or Fake Job Posting Detection Sridevi, Likitha, Chandana & Shamarthi (2024): This study uses a dataset of 17,880 entries and applies classical ML classifiers (K-Nearest Neighbors, SVM, Random Forest, Logistic Regression, Naïve Bayes, AdaBoost) over textual and structured features (job title, company profile, description, industry, salary, etc.). The authors report high classification performance, especially using Random Forest
- Fake Job Detection and Analysis Using Machine Learning and Deep Learning Algorithms Nagarajan et al. (2025): This paper compares traditional ML and deep-learning models on cleaned and pre-processed job-post data. It emphasizes careful data cleaning, feature engineering (unigrams/bigrams), and shows that deep-learning models outperform classical approaches in detecting fake job posts.
- Online Fake Job Advertisement Detection using Machine Learning Techniques (2024): This work uses preprocessing, TF-IDF vectorization, and ensemble classifiers (e.g., Random Forest) to detect fraudulent job ads. It argues for scalability

and automation as a defense against the growing volume of fraudulent postings.

- Predicting Fake Job Posts Using Machine Learning Models Boka (2024): This study applies a variety of classifiers (decision trees, logistic regression, SVM, naïve Bayes, random forest, multilayer perceptron) on a dataset of job ads, using data mining algorithms like Apriori for feature selection before classification. The work demonstrates the feasibility of automated fake-job detection using ML techniques.

7.2 Deep Learning & Multimodal / Advanced Feature Approaches

- Detecting Fake Job Postings Using Bidirectional LSTM Pillai (2023): This paper proposes a Bi-LSTM model over textual and numeric features to distinguish fake from real job ads. The authors report a high performance: 98.71% accuracy and 0.91 ROC AUC, underscoring the potential of sequence models for semantic analysis of job descriptions.
- Detecting Online Recruitment Fraud through Deep Learning 2025 study using CNN-based feature extraction (CNN2D) combined with structured data to classify fraudulent job ads. This moves beyond simple text features to richer learned feature representations, improving classification robustness.
- Visualizing and Clustering Fake Job Postings (2025): This work explores visualization and clustering to uncover patterns common to fake job ads such as industry-targeting trends, role types, and suspicious textual patterns. Though not a classification model, it provides valuable qualitative insights that can guide feature engineering or data sampling strategies.

7.3 Insights from Related Domains: Phishing, Spam & Multimodal Detection

- Advanced BERT and CNN-Based Computational Model for Phishing Detection in Enterprise Systems Gupta et al. (2024): Although focused on phishing email detection rather than job ads, this work demonstrates the power of combining pre-trained language models (BERT) for text feature extraction with CNN-based classification for phishing content detection. The conceptual similarity supports using hybrid textual + deep-learning models for scam detection tasks.
- Multimodal Framework for Phishing Attack Detection and Mitigation Murhej et al. (2025): This study integrates multiple data sources, SMS, email body, URLs, CSS, and images to detect phishing attacks. The use of multimodal inputs (text + structural + visual/URL features) offers a compelling precedent for applying such an approach to job-post fraud detection.
- Spam Detection Using Bidirectional Transformers and Machine Learning Guo et al. (2023): This research employs transformer-based embeddings (e.g., BERT) combined with classical classification methods to detect spam content. Though applied to spam rather than job ads, it demonstrates how advanced NLP embeddings improve detection of malicious or deceptive content, which is relevant for detecting job-ad fraud.

7.4 Summary of Observed Trends and Gaps

- Traditional machine-learning approaches (TF-IDF + classical classifiers) remain popular and often yield high accuracy, but they rely heavily on manual feature engineering.
- Deep-learning models, especially sequence models (e.g., Bi-LSTM) and CNN-based approaches, offer more capacity to learn nuanced semantic or structural patterns without manual feature crafting.
- Multimodal methods, though still more common in phishing or spam detection, show significant promise: they combine textual, structural (metadata), and potentially visual cues to capture complex fraud signals that single-modality models may miss.
- There remains a relative scarcity of research applying full multimodal systems (text + structured metadata + images) to fake job posting detection, representing a potential gap and opportunity for novel contributions.

8 Conclusions

We presented a comprehensive multimodal fraud detection system integrating text, image, and metadata. More importantly, we incorporated explainability techniques deeply into the model pipeline, ensuring transparent, trustworthy predictions. Future extensions include longer training, more metadata fields, adversarial testing, and full multimodal deployment.

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